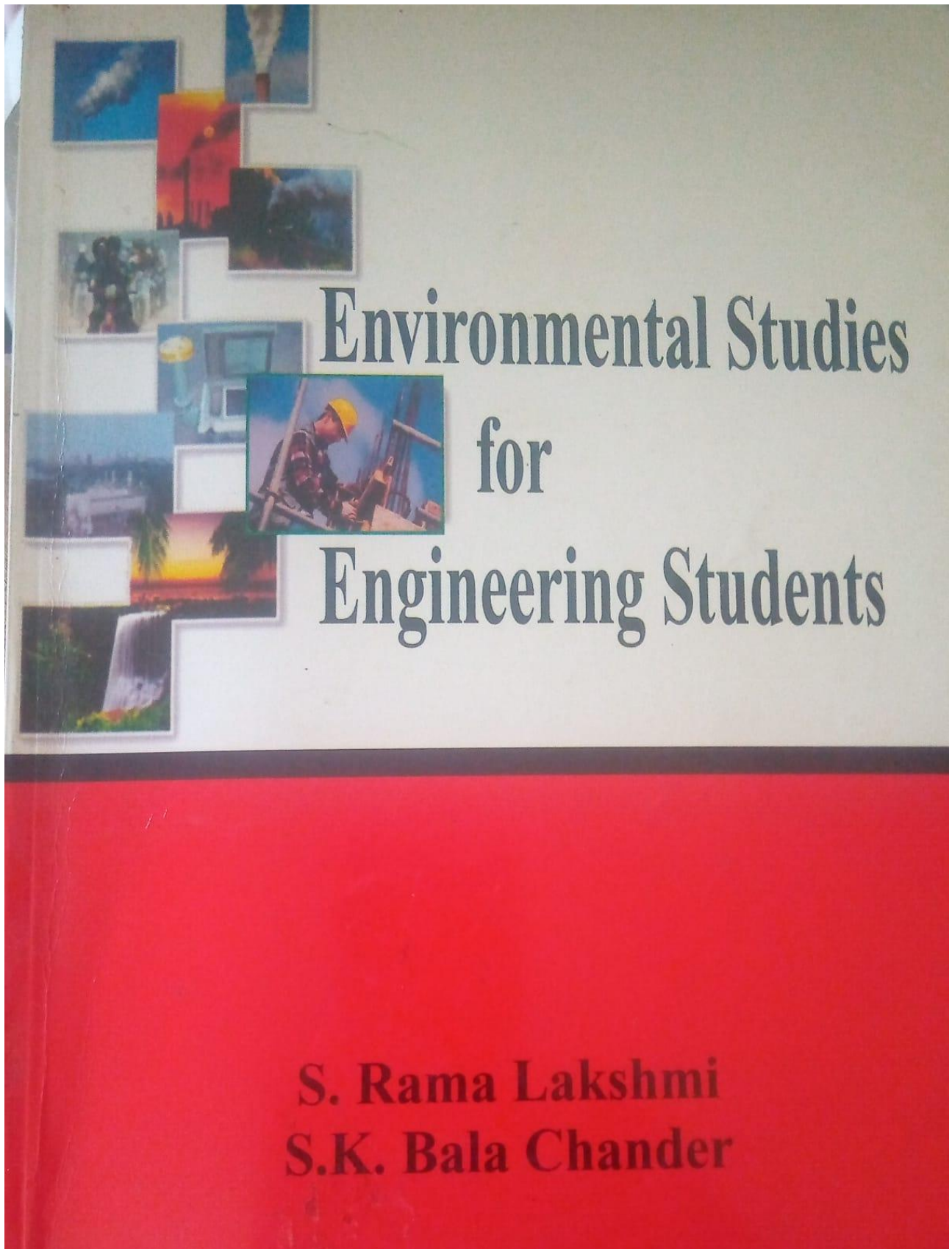




ENVIRONMENTAL STUDIES **For** **ENGINEERING STUDENTS**

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SYLLABUS

**B TECH IN COMPUTER SCIENCE ENGINEERING
IV YEAR 1SEM**

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ENVIRONMENTAL SCIENCES

Objectives:

1. Understanding the importance of ecological balance for sustainable development.
2. Understanding the impacts of developmental activities and mitigation measures.
3. Understanding the environmental policies and regulations

Outcomes:

Based on this course, the Engineering graduate will understand /evaluate / develop technologies on the basis of ecological principles and environmental regulations which in turn help in sustainable development.

UNIT-I :

Ecosystems: Definition, Scope and Importance of ecosystem. Classification, structure and function of an ecosystem, Food chains, food webs and ecological pyramids. Flow of energy, Biogeochemical cycles, Bioaccumulation, Biomagnification, ecosystem value, services and carrying capacity, Field visits.

UNIT-II:

Natural Resources: Classification of Resources: Living and Non-Living resources, **water resources:** use and over utilization of surface and ground water, floods and droughts, Dams: benefits and problems. **Mineral resources:** use and exploitation, environmental effects of extracting and using mineral resources, **Land resources:** Forest resources, **Energy resources:** growing energy needs, renewable and non-renewable energy sources, use of alternate energy source, case studies.

UNIT-III:

Biodiversity and Biotic Resources: Introduction, Definition, genetic, species and ecosystem diversity. Value of biodiversity; consumptive use, productive use, social, ethical, aesthetic and optional values. India as a mega diversity nation, Hot spots of biodiversity. Field visit. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts; conservation of biodiversity: In-Situ and Ex-situ conservation. National Biodiversity act.

UNIT-IV:

Environmental Pollution and Control Technologies: Environmental Pollution: Classification of pollution, **Air Pollution:** Primary and secondary pollutants, Automobile and Industrial pollution, Ambient air quality standards. **Water pollution:** Sources and types of pollution, drinking water quality standards. **Soil Pollution:** Sources and types, Impacts of modern agriculture, degradation of soil. **Noise Pollution:** Sources and Health

hazards, standards, **Solid waste:** Municipal Solid Waste management, composition and characteristics of e-Waste and its management. **Pollution control technologies:** Wastewater Treatment methods: Primary, secondary and Tertiary. Overview of air pollution control technologies, Concepts of bioremediation. **Global Environmental Problems and Global Efforts:** Climate change and impacts on human environment. Ozone depletion and Ozone depleting substances (ODS). Deforestation and desertification. International conventions / Protocols: Earth summit, Kyoto protocol and Montréal Protocol.

UNIT-V:

Environmental Policy, Legislation & EIA: Environmental Protection act, Legal aspects Air Act- 1981, Water Act, Forest Act, Wild life Act, Municipal solid waste management and handling rules, biomedical waste management and handling rules, hazardous waste management and handling rules. EIA: EIA structure, methods of baseline data acquisition. Overview on Impacts of air, water, biological and Socio-economical aspects. Strategies for risk assessment, Concepts of Environmental Management Plan (EMP). **Towards Sustainable Future:** Concept of Sustainable Development, Population and its explosion, Crazy Consumerism, Environmental Education, Urban Sprawl, Human health, Environmental Ethics, Concept of Green Building, Ecological Foot Print, Life Cycle assessment (LCA), Low carbon life style.

SUGGESTED TEXT BOOKS:

- 1 Textbook of Environmental Studies for Undergraduate Courses by Erach Bharucha for University Grants Commission.
- 2 Environmental Studies by R. Rajagopalan, Oxford University Press.

REFERENCE BOOKS:

1. Environmental Science: towards a sustainable future by Richard T.Wright. 2008 PHL Learning Private Ltd. New Delhi.
2. Environmental Engineering and science by Gilbert M.Masters and Wendell P. Ela .2008 PHI Learning Pvt. Ltd.
3. Environmental Science by Daniel B.Botkin & Edward A.Keller, Wiley INDIA edition.
4. Environmental Studies by Anubha Kaushik, 4th Edition, New age international publishers.
5. Text book of Environmental Science and Technology - Dr. M. Anji Reddy 2007, BS Publications.

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UNIT-I

ECOSYSTEMS

1.1. Definition of Ecosystem

‘Eco’ implies the environment,

‘System’ implies an interacting, inter-dependent complex.

Ecology is the study of reciprocal relationship between organisms and their environment.

Ecosystem is the basic functional unit of ecology.

Any unit that includes all the organisms i.e. the communities in a given area, interact with the physical environment so that a flow of energy leads to clearly defined trophic structure, biotic diversity and material cycle (i.e. exchange of materials between living and non-living components) within the system is known as ecological system or ecosystem.

1.1.1. Scope and importance of Ecosystem:

Ecology is the science that needs minimum time and labors for its introduction to a layman. These days ecology has been contributing very much to socio-economic, political and other similar policies of the world. Ecology plays an important role in agriculture (crop rotation, weed control, etc.), management of grasslands, forestry, biological surveys, pest control, fishery biology, and in the conservation of soil, wildlife, forest, water supplies, etc. The international problem of environmental pollution also needs ecological assistance.

1.1.2. Basic concept of ecosystem:

1. Biotic and abiotic components are considered in basic **structural** and **functional** units of natural ecosystems.
2. The components exist varying degrees of +,- or even neutral **interactions among** organisms.
3. The **radiant energy** is trapped by the autotrophic organisms and is transferred as organic molecules to the heterotrophic organisms. This energy flow is uni-directional or non-cyclic.
4. The chemical components of the ecosystem move in defined cycles- **biogeochemical cycles**.
5. Successful growth of the organism is governed by **limiting factors**.
6. Under natural conditions, different kinds of population undergo **succession**.
7. Then come to the probabilities of **disruption** and **exploitation** of ecosphere. As a result of natural condition or activities of man, species diversity of an ecosystem is reduced. It leads to a set back to the state of development and reduction in the stability of the ecosystem.

1.1.3. Classification of Ecosystems:

The kind of organisms which can live in a particular ecosystem depends upon their physical and metabolic adaptations to the environment of that place/ ecosystem. The set of ecosystems is called a biome.

In the biosphere, there are – natural and artificial biomes.

1. Natural ecosystems (biomes):

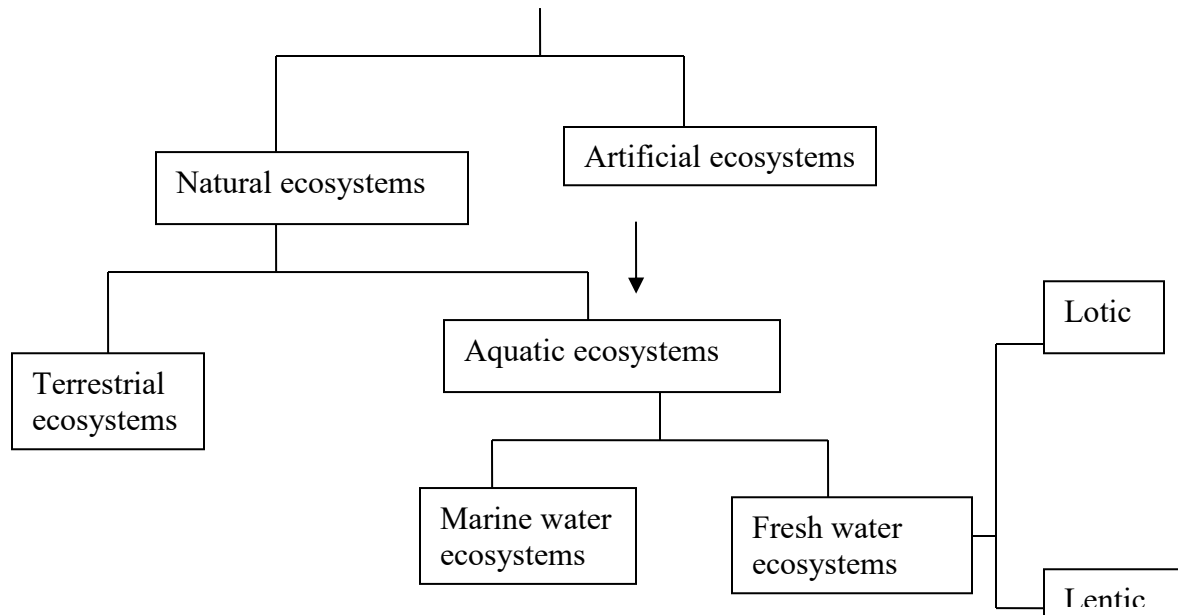
Naturally ecosystems operate by themselves under natural conditions without any interference by man. On the basis of particular type of habitat, they are further subdivided as:

a. Terrestrial biomes (ecosystems): They are often defined by the vegetation types that dominate the community. This types of vegetation affect the climate and soil structure and thus characterize the particular biome. Terrestrial vegetation has a rapid exchange of oxygen, water and carbon dioxide. The CO₂ concentration is affected by terrestrial vegetation seasonally and annually. Terrestrial biomes include tropical rain forests, grasslands, deserts, cultivated lands etc.

b. Aquatic biomes (ecosystems): They fall into two categories, viz., freshwater and marine water. Fresh water biomes may be lotic (running-water) such as streams, rivers and springs; or lentic (standing- water) such as lakes, ponds, and swamps. Whereas, marine biomes include deep sea and oceans.

2. Artificial ecosystems (biomes): They are maintained artificially by man. A pond constructed as part of a waste water treatment plant is an example of artificial ecosystem. And other examples are croplands, aquarium, aquaculture ponds etc.

Ecosystems



1.2. Structure and function of an Ecosystem/ organization of an ecosystem:

The structure and function are the two major aspects of an ecosystem.

Structure means:

- i. the composition of biological community including species, numbers, biomass, life history and distribution in space etc.
- ii. quantity and distribution of the non-living material, such as nutrients, water, soil etc.
- iii. range or gradient of conditions of existence, such as temperature, light etc.

Function means:

- i. the rate of biological energy flow i.e, the production and respiration rates of the community.
- ii. rate of materials or nutrient cycles, and
- iii. biological or ecological regulation including both regulation of organisms by environment (photoperiodism etc) and regulation of environment by the organism (nitrogen fixing organisms etc)

1.2.1. Structure of an Ecosystem:

Ecosystem has two major components:

i. Abiotic (non-living) component:

It includes Chemical factors:

1. Amount of inorganic substances as P,S,C,N,H etc. involved in material cycles. The amount of these inorganic substances present at any given time in an ecosystem, is called as 'standing state' or 'standing quality'.

2. amount and distribution of inorganic chemicals, such as chlorophyll, etc. organic materials such as proteins, carbohydrates, lipids etc. present either in the biomass or in the environment i.e. 'biochemical structure' that link the biotic and abiotic components of the ecosystem.

Physical or climatic factors such as soil, temperature, light and water.

ii. Biotic(living) components:

Trophic (nutritional) stand point, an ecosystem has two components:

1. Autotrophic component:

Autotrophs presence of sunlight using simple inorganic substances and built up of complex substances. The component is constituted by green plants, including photosynthetic bacteria.

Members of the autotrophic components are known as 'producers'.

2. Heterotrophic components:

In which utilization, rearrangement and decomposition of complex materials. The organisms involved are known as 'consumers', as they consume the matter built up by the producers.

These consumers are further categorized as-

a. Macro consumers: are further categorized as-

i. Primary consumers(Herbivores): they feeding on grass, plant leaves etc. are mainly such grazing animals as cows, buffaloes, deer's, sheep, rabbit, mouse etc. Besides that there are also present some insects, termites, millipedes etc. that feed on the leaves of grasses.

ii. Secondary consumers (carnivores): These are the carnivores feeding on herbivores. These include the animals like fox, jackal, snakes, frogs, lizards, birds etc.

iii. Tertiary consumers (carnivores/omnivores): These are the carnivores like lion, tiger etc. that eat secondary consumers.

b. Micro consumers: These are popularly known as 'decomposers', include bacteria, actinomycetes and fungi.

They breakdown complex compounds of dead or living protoplasm, absorb some of the decomposition or breakdown products and release inorganic nutrients into environment, making them again to autotrophs.

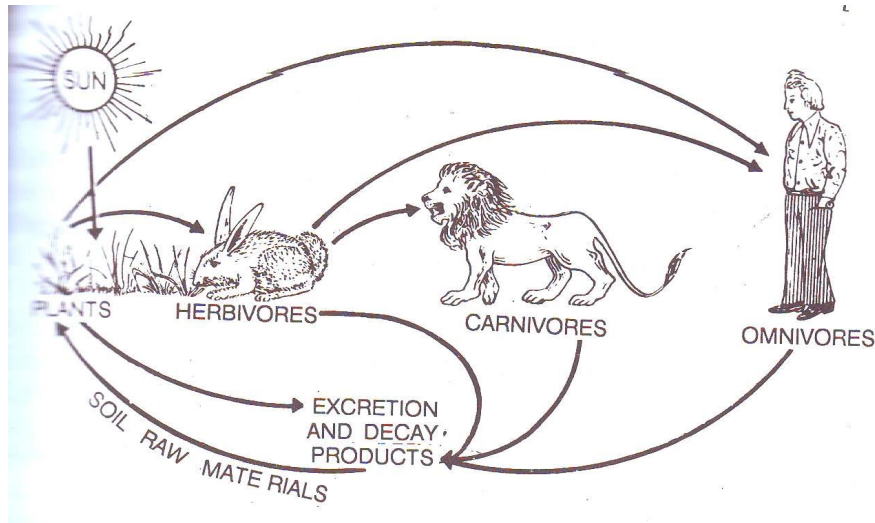


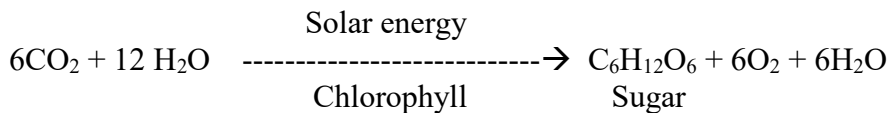
Fig: A generalized scheme of nutritional relationships among different biotic components of an ecosystem. .

1.2.2. Function of an ecosystem:

The function of ecosystem is related to energy flow and material cycles within and outside of the system.

Fixation of energy: utilizes solar energy (2 calories/cm²/minute) in photosynthesis.

Photosynthesis by autotrophs.



This sugar has several destinations.

1. Storage : sugar can be converted into an inert, energy rich organic substance as starch.
2. Sugar combine with other molecules to form specialized carbohydrates as cellulose.
3. Sugar can combine with N, P, S, to build complex molecules like proteins, nucleic acids, pigments, hormones etc.
4. Oxidation of sugar yields usable energy which is released into the ecosystem.



1.2.2.1. Energy flow in an ecosystem:

Energy flow is the movement of energy through an ecosystem- from the external environment through a series of organisms and back to the external environment. It is one of the most fundamental processes that is common to all the ecosystems. Energy flow in

an ecosystem, in fact, tells us about the very ecosystem, the energy flow provides a foundation for life and thus imposes a limit on the abundance and richness of life. The behavior of energy in ecosystem can be treated energy flow due to unidirectional flow of energy.

From energetic point of view it is essential to understand for an ecosystem:

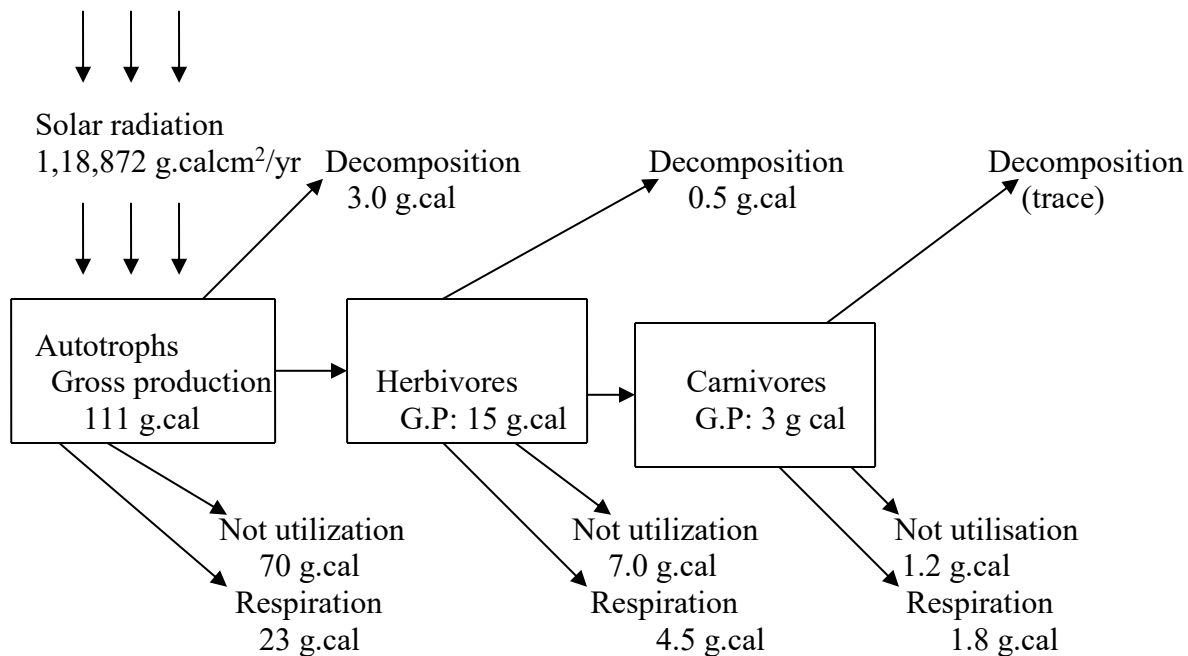
1. the efficiency of the producers in absorption and conversion of solar energy.
2. the use of this converted chemical form of energy by the consumers,
3. the total input of energy in form of food and its efficiency of assimilation,
4. the loss through respiration, heat, excretion, etc. and
5. the gross net production.

Energy flow is two types:

1. Single channel flow
2. Y- shaped energy flow

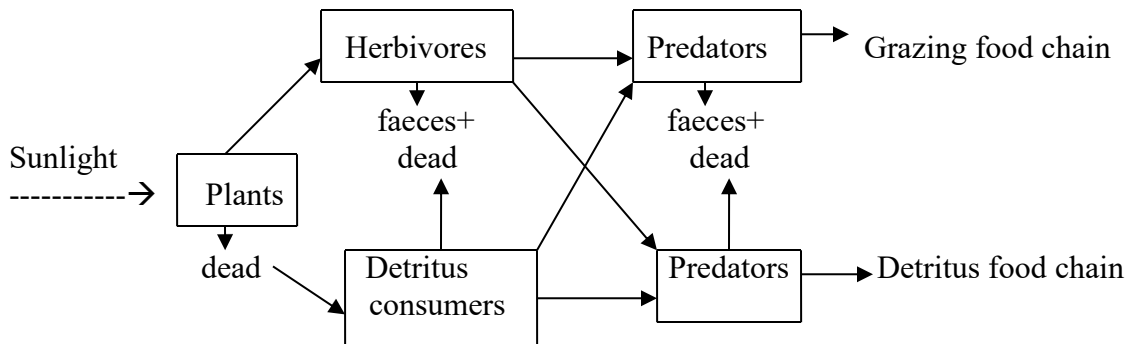
These two are basic food chains in any ecosystem.

1. Single channel energy flow: the grazing food chain beginning with green plant base going to herbivores and then to carnivores.



2. Y- shaped energy flow:

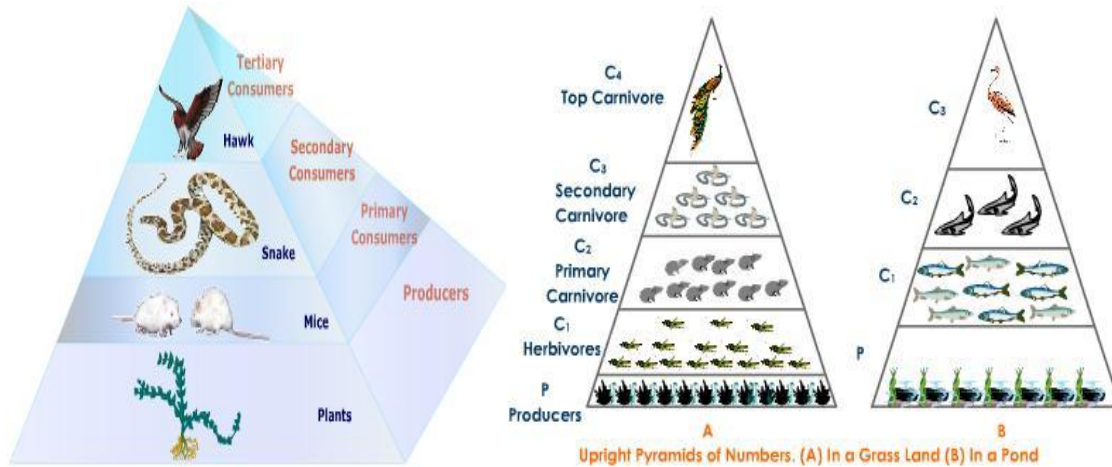
The detritus food chain beginning with dead organic matter acted by microbes, then passing to detritivores and their consumers (predators).



1.2.2.2. Ecological Pyramids:

The interaction of the food chain phenomenon (i.e. energy loss at each transfer) and the size metabolism relationship results in communities, having definite structure. The graphical representation of the trophic structure and also trophic function is called 'ecological pyramids'. In ecological pyramid, the first or producer level forms the base and successive levels form the tiers which make up the apex. In simple words, if we arrange the organisms in a food chain according to trophic levels, they often form pyramid. The ecological pyramids are of three types, viz., the pyramid of numbers, the pyramid of biomass and the pyramid of energy.

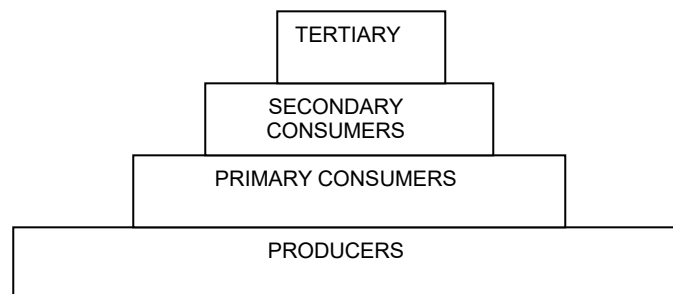
1. The pyramid of numbers: In this the numbers of individual organisms at different trophic levels in an ecosystem are depicted. The total numbers of individual organisms at producer levels (first trophic level) form the base of the numbers pyramid, and the population of primary consumers, secondary consumers, tertiary consumers, and so on forms the successive tiers of the pyramid of numbers. The length of the bar, at different levels (tiers), represents the number (population) of organisms at that particular trophic level, by using convenient scale.



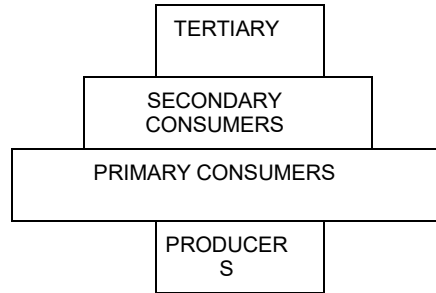
Pyramid of number in grassland

The number of pyramid may be inverted or partly so, that is, the base may be smaller than one or more of the upper tiers.

2. Pyramid of biomass: It is based on the total dry weight, calorific value, or any other measure of the total amount of living material. In this, the total amount of living material (say in gms. Dry wt. per. Sq.mt) in organisms at different levels is depicted. When larger weight of producers support a smaller weight of consumers an upright pyramid results. eg forest ecosystem. When smaller weight of producers supports larger weight of consumers an inverted pyramid of biomass is formed eg. aquatic ecosystem.

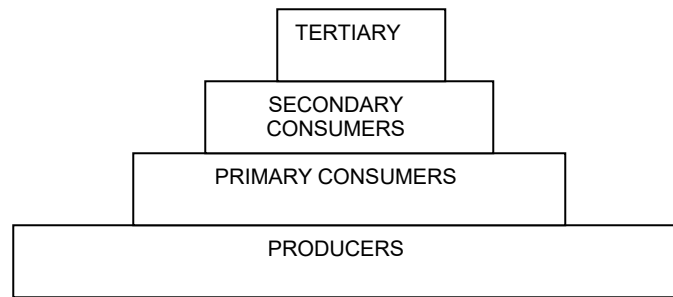


Forest (upright)



Pond (inverted)

3. The pyramid of energy: In this, the rate of energy flow and/or productivity at successive trophic levels is shown. The pyramid of energy is drawn after taking into consideration the total quantity of energy utilized by the trophic levels in an ecosystem over a period of time. As the quantity of energy available for utilization in successive trophic levels is always less because there is loss of energy in each transfer, the energy pyramid will always be upright.



Pyramid of energy in all ecosystems

The energy pyramid always takes a true upright pyramid shape, provided all sources of food energy in the system are considered, because less food energy is available to the top trophic level than is available to preceding level.

1.2.2.3. Food chains:

In any food chain, energy flows from primary producers to primary consumers (herbivores), from primary consumers to secondary consumers (carnivores), and from secondary consumers to tertiary consumers (carnivores/omnivores) and so on.

This simple chain of 'eating and being eaten away' is known as food chain.

The length of the chain denotes the complexity of ecosystem.

Ex: Grass → Insects → Frog → Snake (in a grassland)

Two types of food chains:

1. Grazing food chain: starts from living green plants.

Ex. Grass → Insects → Frog → Snake (in a grassland)

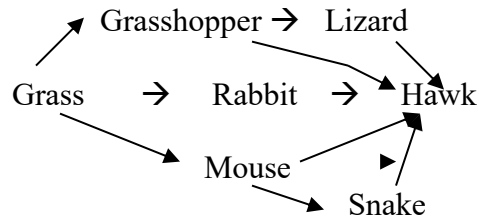
2. Detritus food chain: This food chain goes from dead matter into micro organisms.
Ex. Mangrove leaves → Micro organisms (bacteria, fungi) → insect, larvae → Amphipods → Small game fish → fish eating birds.

1.2.2.4. Food web:

Food chains in natural conditions never operate as isolated sequences, but are interconnected with each other forming some sort of inter locking pattern, which is referred to as a food web.

Ex. Grassland ecosystem: in this five linear food chains are interlinking form food web.

1. Grass → Grasshopper → Hawk
2. Grass → Grasshopper → Lizard → Hawk
3. Grass → Rabbit → Hawk
4. Grass → Mouse → Hawk
5. Grass → Mouse → Snake → Hawk



1.3. Types of Ecosystems

1. Aquatic ecosystems (pond, streams, lake, oceans, estuaries etc.)
2. Terrestrial ecosystems (forest, grassland, desert etc.)

Aquatic ecosystems ex: Pond:

Basic components in the pond ecosystem are:

1. Abiotic components: are heat, light, P^H value of water, and basic inorganic and organic compounds, such as water itself, carbon dioxide, oxygen, calcium, nitrogen, phosphates, amino acids, humic acid etc. The amount of the mineral matter present at any time in the physical environment of the pond is 'standing state'.

2. Biotic components: The various organisms that constitute the biotic component are as follows:

a. Producers: These are autotrophic, green plants and some photosynthetic bacteria. The producers fix radiant energy and with the help of minerals derived from the water and mud, they manufacture complex organic substances as carbohydrates, proteins, lipids etc.

Producers are the following types:

i. Macrophytes: These are mainly rooted larger plants which include partly or completely submerged, floating and emergent hydrophytes.

Ex. Trapa, typha, hydrilla, vallisneria, lemna etc.

ii. Phytoplanktons: these are minute, floating or suspended lower plants, majority of them are such filamentous algae as Zygnema, Volvox, Spirulina, Diatoms etc.

b. Consumers: They are heterotrophs which depend for their nutrition on the organic food manufactured by producers, the green plants.

Most of the consumers are herbivores.

Consumers are further divided into-

i. Primary consumers (herbivores): These are feeding directly on living plants (producers) or plant remains. These may be large as well as minute in size.

1. Benthos: those are the animals associated with living plants and those are bottom forms which feed upon the plant remains living at the bottom of ponds.

Ex. Buffaloes, cows etc.

2. Zooplanktons: they feed chiefly on phytoplanktons

Ex. Euglena, Coleps, Dileptus etc.

ii. Secondary consumers (carnivores): These are feeding on the primary consumers. These are chiefly insects and fish.

iii. Tertiary consumers (carnivores/omnivores) These are large fish as game fish that feeding on the smaller fish.

c. Decomposers : They are also known as ‘micro consumers’.

They bring about the decomposition of complex dead organic matter of both producers (plants) as well as macro consumers (animals) to simple forms. Thus they play an important role in the return of mineral elements again to the medium of the pond.

Ex. Cephalosporium, Aspergillus, Rhizopus, etc. are most common decomposers in water mud of the pond.

Terrestrial Ecosystems ex: 1. Forest.

The different components of a forest ecosystem, like others, are as follows-

1. Abiotic components:

These are the inorganic as well as organic substances present in the soil and atmosphere. In addition to the minerals present in forests are dead organic debris.

2. Biotic components:

The living organisms present in the food chain occur in the following order.

i. Producers: These are mainly trees. The trees are of different kinds depending upon the kind of the forest formation developing in that climate. Besides trees, there are also present shrubs and a ground vegetation. In these forests, dominant members of the flora, the producers, are such trees as Tectona grandis, Butea frondosa etc.

ii. Consumers: These are as follows-

a. Primary consumers: These are herbivores that include the animals feeding on tree leaves as ants, flies, beetles, leafhoppers, bugs and spiders etc. and larger animals grazing on shoots or fruits of the producers, the elephants, deers, moles, squirrels, shrews, flying foxes etc.

b. Secondary consumers: These are carnivores like snakes, birds, lizards, fox etc. feeding on the herbivores.

c. Tertiary consumers: These are top carnivores like lion, tiger etc. that eat carnivores of secondary consumers level.

iii. Decomposers: These are wide variety of micro organisms including fungi (species of *Aspergillus*, *Coprinus*, *Alternaria* etc.), Bacteria (Species of *Bacillus*, *Clostridium*, *Pseudomonas* etc) and actinomycetes, like species of *Streptomyces* etc.

Rate of decomposition in tropical and sub tropical forests is more rapid than that in the temperate ones.

Terrestrial ecosystem ex2: Grassland ecosystem:

This is a type of terrestrial ecosystem.

The various components of a grassland ecosystem are as follows:

1. Abiotic components: These are nutrients present in soil and the aerial environment. Thus the elements like C,H,O,N,P,S etc. are supplied by CO₂, H₂O, nitrates, phosphates and sulfates etc. present in air and soil of the area.

2. Biotic components: These may be categorized as-

a. Producers: They are mainly grasses as species of *Dichanthium*, *Cynodon*, *Desmodium* etc. Besides them a few forbs and shrubs also contribute to primary production.

b. Consumers: These occur in the following sequence-

i. Primary consumers: The herbivores feeding on grasses are mainly such grazing animals as cows, buffaloes, deers, sheep, rabbit etc. Besides them, these are also present some insects as *Cicincella*, *Coccinella*, some millipeds etc. that feeding on the leaves of grass.

ii. Secondary consumers: These are carnivores feeding on herbivores. These include the animals like fox, jackal, snakes, frogs, lizards, birds etc.

iii. Decomposers: The microbes active in the decay of dead organic matter of deferent forms of higher life are fungi, as species of *Mucor*, *Aspergillus*, *Penicillium* etc.

1.4. Biogeochemical cycles:

The cyclic flow of nutrients between the biotic and abiotic components is known as nutrient cycle or biogeochemical cycles.

The nutrients enter into producers and move through the food chain and ultimately reach the consumer. The bound nutrients of the consumers, after death, are decomposed and converted into inorganic substances, which are readily used up by the plants and again the cycle starts.

The major nutrients like C,H,O,N, P and S are cycled again and again between biotic and abiotic components of the ecosystem.

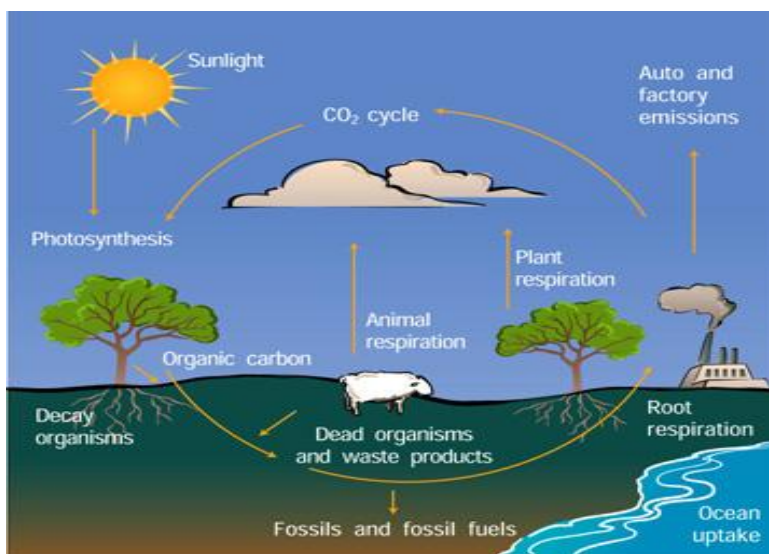
Carbon Cycle:

Carbon is the basic component in all the organic compounds. The carbon is present in all biotic components in different forms as food.

Carbon is present in the atmosphere as CO_2 . The CO_2 is taken up by the green plants as a raw material for photosynthesis of different food. This food moves through food chain, finally the carbon present in the dead matter is returned to the atmosphere as CO_2 by microorganisms.

Sources of CO_2 in atmosphere

1. During respiration, plants and animals liberates CO_2 in the atmosphere.
2. Combustion of fuels releases CO_2
3. Volcanic eruptions also releases CO_2



Nitrogen cycle

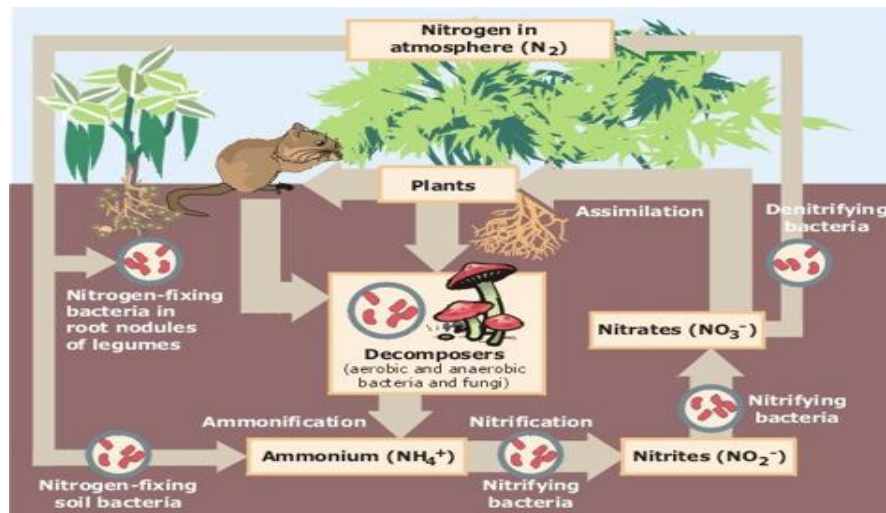
Nitrogen is present in the atmosphere as N_2 in large amounts (78%). The nitrogen is present in all biotic components in different forms as food.

Ex. Proteins, vitamins, amino acids etc.

The nitrogen form in the atmosphere is taken up by the green plants as a raw material for biosynthesis of different foods (Proteins, vitamins, amino acids etc) and used in metabolism. This food moves through the food chain. After death of the plants and animals, the organic nitrogen in dead tissues is decomposed by several micro organisms (ammonifying and nitrifying bacteria) into ammonia, nitrites and nitrates, which are again used by the plants. Some bacteria convert nitrates into molecular nitrogen, which is again released back into atmosphere and the cycle goes on.

Nitrification: The conversion of ammonia into nitrates is termed as nitrification. This is brought about by nitrifying bacteria. Ex. Nitrobacter, Nitrasomonas.

Denitrification: The conversion of nitrates into nitrogen is termed as denitrification. This process is brought about by denitrifying bacteria. Ex. Pseudomonas, Fluorescence.



Hydrological Cycle:

Movement of water in a cyclic manner is known as hydrological cycle.

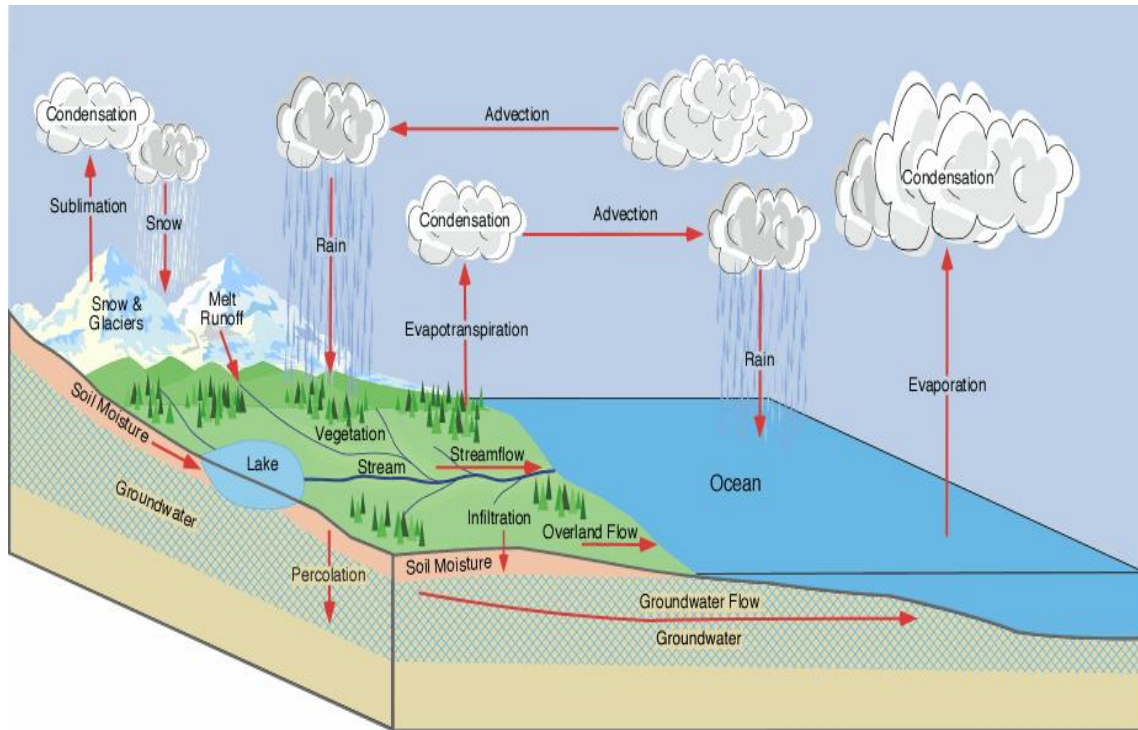
Hydrological cycle involves the following steps

1. Evaporation
2. Condensation and precipitation
3. Transpiration and respiration.

1. Evaporation: Heat energy from the sun constantly causes evaporation from all the water surfaces. Oceans, rivers, streams, lakes, ponds and the surfaces of terrestrial organisms lose water due to evaporation.

2. Condensation and precipitation: Precipitation occurs due to the condensation of water from a gaseous state in the atmosphere and falls to earth. Once water condenses, it is pulled into the ground by gravity. Gravity continues to operate either pulling the water underground or across the surface (surface run-off). In both the cases gravity continues to pull the water lower and lower until it reaches the oceans.

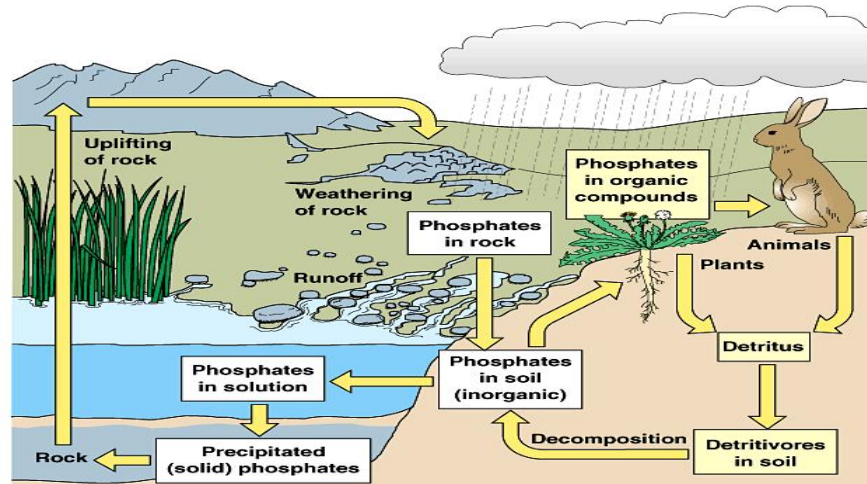
3. Transpiration and respiration: Organisms play an important role in the water cycle. Plants use the soil water for photosynthesis and to transport materials within them. Much of the water, they absorb through their roots, is lost to the atmosphere from the leaves. This process is known as transpiration. In both animals and plants, the breakdown of sugars to produce energy (known as respiration) with the liberation of by products CO₂ and Water



Phosphorous Cycle:

Phosphorous is mainly present in the rocks and fossils. The Phosphorous is present in all biotic component in different forms. Ex. Bones, teeth's etc.

Phosphate rocks is excavated by man for using it as a fertilizers. Farmers use excess of fertilizers for the crops. The excess phosphate fertilizers move with the surface run-off, reaches the oceans and are lost into the deep sediments. Sea birds eat sea- fishes, which are Phosphorous rich and the excreta of the birds return the Phosphorous to the land. Thus the sea birds are playing an important role in Phosphorous cycling. Animals and plants use these dissolved phosphates during the biosynthesis.



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1.5. Homeostasis:

Homeo= same; stasis= standing.

Is the tendency for the biological system to resist change and to remain in a state of possible equilibrium.

1.6. Cybernetics:

The ecosystems are capable of self- maintenance and self- regulations as are their component populations and organisms. This is the science of control mechanisms and communications.

1.7. Biomagnification/ bioamplification/ food chain concentration:

Some of the most toxic biocides are DDT, BHC, Chlordane, Heptachlor, Methoxychlor, Adrin, Endrin and Polychlorinated Biphenyles (PCB's). Indiscriminate use of the biocides could make them an integral part of our biological, geological and chemical cycles of the earth. Measurable amounts of DDT residues may be found in air, soil, water and at several thousands of kms from the point where it had originally entered the ecosystem. For instance of DDT enters a pond, lake, it is taken as such by the plants of the pond, then reaches to Zooplanktons feeding on plants, then to minnows eating the zooplanktons, then to fish which eats the minnows and finally in the body of birds who eat the fish.

Not only DDT as such in its original form keeps on moving from water to different living components of the pond system but more threatening is that DDT concentration continuously increases in successive trophic levels (various form of living organisms) in a food chain. This phenomenon is known as biological magnification or biological amplification. This is the reason why our food grains as wheat, rice and vegetables, fruits today containing various amounts of pesticide residues which have become their integral part. They cannot be removed by washing or other means.

Effects: Kidney malfunctioning, excess of amino acids in blood and urine, electroencephalogram abnormalities of brain tissue, blood abnormalities etc.

Ex: Phytoplanktons $\xrightarrow{1\text{ppm}}$ Zooplanktons $\xrightarrow{2\text{ppm}}$ Big fish $\xrightarrow{5\text{ppm}}$ Man/ birds $\xrightarrow{10\text{ppm}}$

1.8. Bio- accumulation:

Increase of pesticide concentration in single animal cell is called as bio-accumulation. Pesticide concentration increased in single animal by daily intake of contaminated (by pesticides) food like vegetables, water etc.

1.9. Ecosystems services:

Human kind benefits from a multitude of resources and processes that are supplied by natural ecosystems. Collectively, these benefits are known as ecosystem services. Experts currently recognize five categories of ecosystem services. The following lists represent samples of each:

Provisioning services:

- Food (including sea food), crops, wild foods and species.
- Water
- Pharmaceuticals, bio-chemicals and industrial products.
- Energy (hydropower, biomass fuels)

Regulating services:

- Regulate disease carrying organisms.
- Contribute to climate stability.
- Generate and preserve soils and renew their fertility.
- Moderate weather extremes and their impacts.
- Carbon sequestration and climate regulation
- Waste decomposition and detoxification.
- Purification of water and air.
- Crop pollination.
- Pest and disease control.

Protective services:

- Protect stream and river channels and coastal shores from erosion.
- Protect people from the sun's harmful UV rays.

Supporting services:

- Nutrient dispersal and cycling.
- Seed dispersal.
- Primary production.

Cultural services:

- Cultural, intellectual and spiritual inspiration.
- Recreational experiences (including ecotourism)
- Scientific discovery.

1.10. Value of Ecosystems:

Values are divided into six types.

1. Consumptive use value:

Value assigned to consumption of resources by the people.

Ex. Utilization of timber, food, fuel wood, fodder by local communities.

2. Productive use value:

This value comprises of marketable goods.

A variety of industries like pharmaceuticals, are highly dependent on identifying compounds of great economic value from the wide variety of wild species of plants located in undisturbed natural forests. This is called 'biological prospecting'.

Meat, skins, sandal wood etc. sold in national and international markets to earn money.

3. Social values:

Refers to the manner in which the bio-resources are used to the society. These values are associated with the social life, religion and spiritual aspects of the people.

Ex: Holy plants: Tulsi, lotus etc.

Holy animals: Cow, Snake, Bull, Peacock etc.

4. Ethical and moral values:

Our rich heritage teaches us to worship plants, animals, rivers and mountains. The ethical value means that a species may or may not be used, but its existence in nature gives us pleasure.

Ex: The river Ganga is considered as holy river.

5. Aesthetic Value:

The most important aesthetic value of biodiversity is eco-tourism. [means people from far place spend a lot of time and money to visit the beautiful areas, where they can enjoy the aesthetic value of biodiversity]

The pleasant music of wild birds, color of butterflies, color of flowers etc. are very important for their aesthetic value.

6. Option value:

The optional values of biodiversity suggest that any species may be proved to be a valuable species after some day.

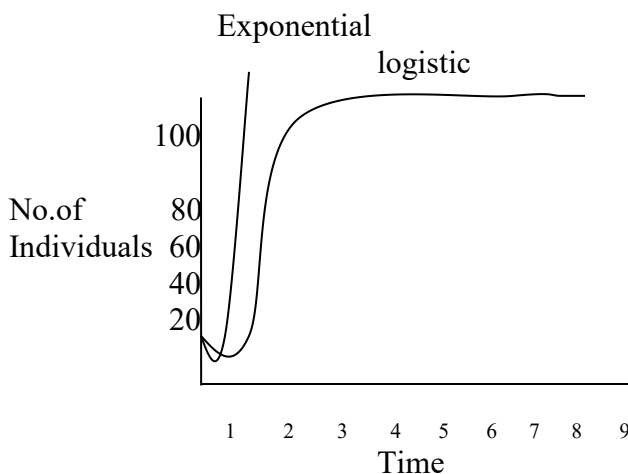
Ex. The growing biotechnology field is searching a species for causing the diseases of cancer and AIDS.

1.11. Carrying Capacity of Ecosystems

The carrying capacity means” how much it can hold”.

Most populations do not show exponential increase because their environment prevents this. In an ecosystem, other species present a variety of abiotic conditions etc. act in a way that a given population will have an upper size limit that cannot be exceeded. Such a limit is called carrying capacity of the environment for that population. Carrying capacity is determined by the availability of resource as food and space.

The logistic equation was devised by Verhulst in 1838 to describe population growth with an upper limit. Suppose, environment has a carrying capacity (K) for a particular population then the logistic equation assumes that the intrinsic rate of natural increase (r) is progressively reduced as population size increases toward that carrying capacity. If there are 10 individuals already present (N=10) and K = 100 individuals, the remaining resource can support a further $K - N = 90$ individuals and the proportion of remaining is $K-N/K = 0.9$



As population density increases, birth rate often decreases and death rate typically increases. The difference between the birth rate and the death rate is the “natural increase”.

The carrying capacity could support a positive natural increase, or could require a negative natural increase.

Thus, the carrying capacity is the number of individuals an environment can support without significant negative impacts to the given organism and its environment.

Below carrying capacity, populations typically increase, while above, they typically decrease.

A factor that keeps population size at equilibrium is known as a regulating factor.

Population size decreases above carrying capacity due to a range of factors depending on the species concerned, but can include insufficient space, food supply, or sunlight.

The carrying capacity of an environment may vary for different species and may change overtime due to variety of factors, including: food availability, water supply, environmental conditions and living space.

The carrying capacity will depend on many factors, such as the nutrient content of the soil, the precipitation (rain, snow and so on), temperature, and light level.

1.12. Visit to a local area to document environmental assets:

Visit may be planned to any nearby river, forest, grassland, hill or mountain, depending upon easy access and importance. Write a report based on your observations and understanding about various aspects of environment. This unit provides the required information for the study and for arriving at some important conclusions about the system.

Study of Food Chains and Webs

Overview:

Visit nearby any ecosystem (parks, grassland etc.) and prepare food chain or food web.

Resources:

- A4 size papers
- research materials
- access to websites

Preparation:

Students complete the food chains and webs worksheet based on one animal found or observed in Ecosystem. Students record information on the selected animal and information about where it lives, what it eats, what its predators are and how it protects itself. This completed food chain and food web helps to demonstrate the interconnectedness of living things in the ecosystem. Discuss:

- what would happen if the tree were removed?
- are any animals eaten more than others?
- why do plants and animals need each other?

Study of River environment

1. Back ground data:

Note down the name of the river, its place of origin and its course or route. Find out whether the river is perennial or seasonal in nature.

2. Water quality observations:

a. Note down whether the water of the river is clear or turbid.

* If it is clear, what do you expect? Penetration of light into water would be more, therefore green aquatic plants will be growing better. The primary productivity will be high.

* If turbid, how would it affect the primary productivity of the river?

b. Note down the temperature of water with a thermometer. Also note the temperature of the air.

* If temperature is more than 5°C than ambient air temperature, what can be the reason?

* Write the probable impacts of thermal pollution on aquatic life.

c. Observe color of water, whether it was dark colored or greasy or foamy. If yes find out the reason.

d. Determine the P^H of water using a P^H- meter.

3. Observations on aquatic life:

Look for different types of life forms. And draw a food-web diagram that would be present in the river.

4. Uses: Prepare the list of the uses for river water.

5. Human Impacts: What are the major impacts caused by surrounding area human beings on the river? Is there any major incident on aquatic life ex. Massive fish death, etc. ? Try to interpret the same.

Review Questions:

1. Define ecosystem and write a detailed note on concepts of ecosystem.

2. Elaborate the Structure and functions of ecosystem.

3. Briefly explain the types of ecosystem.

4. Explain energy flow in ecosystems.

5. Explain in brief- food web, food chains, Ecological pyramids.

6. Explain the characteristics and function of grassland ecosystem.

7. Explain detail cyclic flow of nutrients between biotic and abiotic components with the help of neat cycles.

8. Write a detailed note on ecosystems services and values.

9. Write a note on carrying capacity of ecosystems.

10. Write a report on river and forest ecosystem field visit.

Fill in the blanks

1. The term ecosystem was coined by **Earnest Haeckel**
2. The organisms who feed directly on producers are called **Herbivores**
3. Chemosynthetic organisms can produce organic matter through oxidation of **Chemicals** in the absence of sun light.
4. The sequence of eating and being eaten in an ecosystem is called as **food chain**
5. Biomagnification of the pesticide **thousand times** in an food chain resulted in the thinning of shells in birds eggs.
6. Pyramid of **energy** is always upright.
7. Movement of nutrients in an ecosystem is cyclic while flow of energy is **unidirection**
8. Guano deposits on the coasts of Peru are rich in the nutrient **Phosphorus**
9. The inherent property of all living organisms to resist change is called **homeostasis**
10. The ultimate stable and culminating community during succession is called as **climax**
11. The biggest flower in the plant kingdom is **Rafflesia**
12. The thick layer of ice found frozen under the soil surface throughout the year is called **Permafrost**

Choose the correct answer

1. The organisms which feed on dead organisms, wastes of living organisms are called
a) Chemotrophs b) Carnivores c) Detritivores d) Decomposers [c]
2. The progressive accumulation of some non-biodegradable chemicals through the food chains is known as [b]
a) Ecological balance b) Biological magnification
c) Trophic structure d) Bio-degradation
3. Gross primary productivity is the highest in [c]
a) Open oceans b) Grasslands
c) Wet tropical forests d) Agro ecosystems
4. The type of succession occurring on a base rock is called [b]
a) Halosere b) Lithosere c) Hydrosere d) None of these

5. The tropical grass lands in Africa with tall grasses scattered with shrubs or stunted trees are called [a]
a) Savannas b) Pampas c) Steppes d) Prairies
6. The darker zone in lakes where light penetration is negligible is called [c]
a) Littoral zone b) Limnetic zone c) Profundal zone d) Euphotic zone
7. The over nourished lakes with 'algal blooms' are called [a]
a) Eutrophic b) Oligotrophic c) Dystrophic d) Meromictic
8. Estuaries have the following characteristics [b]
a) Fresh and salt-water b) Rich biodiversity
c) High productivity d) all of these.
9. Valuable, practical services that help to preserve ecosystems performed by nature are called [b]
a) biosphere balancing b) ecosystem services
c) biological control d) the greenhouse effect
10. Which of the following is *not* one of the major environment problems resulting From human interference in the nitrogen cycle? [a]
a) Stratospheric ozone depletion b) Increased acid rain
c) Eutrophication d) Nitrous oxide release increases global warming
11. Which of the following is *not* a major positive feedback mechanism in which the activity of humans to increase global climate temperatures leads to an even further increase? [a]
a) Global warming causes increased rainfall, plant growth and photosynthesis
b) Tropical deforestation causes warming and drying so that remaining forests begin to decline
c) Global warming causes increased CO₂ release from biomass decomposition
d) None of the above
12. The over nourished lakes with 'algal blooms' are called [a]
a) Eutrophic b) Oligotrophic c) Dystrophic d) Meromictic
13. The organisms which feed on dead organisms, waste of living organisms are Called [c]
a) Chemotrophs b) Carnivores c) Detrivores d) Decomposers

Write true are false

1. An ecosystem is a group of biotic communities interacting with one another but without exchanging energy and matter with the non-living environment. **False**
2. Detritivores are also known as saprotrophs. **True**
3. Biotic and abiotic components of an ecosystem influence each other are linked through energy flow and matter cycling. **True**
4. Food webs provided less stability to an ecosystem as compared to linear food chain. **False**
5. Double or Y-shaped energy flow model shows flow of energy through both grazing and detritus food chain. **True**
6. Microorganisms play a crucial role in cycling of nitrogen. **True**
7. The major reservoir of phosphorus is in the atmosphere. **False**
8. Net primary productivity is always more than the gross primary productivity. **False**
9. Wave currents bring nutrients in the estuaries which help in boosting their primary productivity. **True**
10. If we want to maintain the ecological balance, we should try to contribute to positive feedback mechanisms. **True**
11. Positive feed back mechanisms tend to take the system away from its optimal conditions. **True**
12. During ecological succession there is an orderly change in the community structure whereas the physical structure whereas the physical structure remains constant. **True**
13. Landslides, drought, frost or volcanoes can result in development of a bare area for succession to start. **True**
14. Orchids are epiphytes found in abundance in tropical rain forests, known for their beautiful flowers. **True**
15. Abyssal zone of oceans receive no sunlight and constitute an incomplete ecosystem. **True.**